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Black et al.

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(54) **DEVICES AND METHODS FOR ENCOURAGING NASAL BREATHING**

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A61M 16/20 (2006.01)
A61F 5/37 (2006.01)
(Continued)

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(Continued)

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Primary Examiner — Joseph D. Boecker

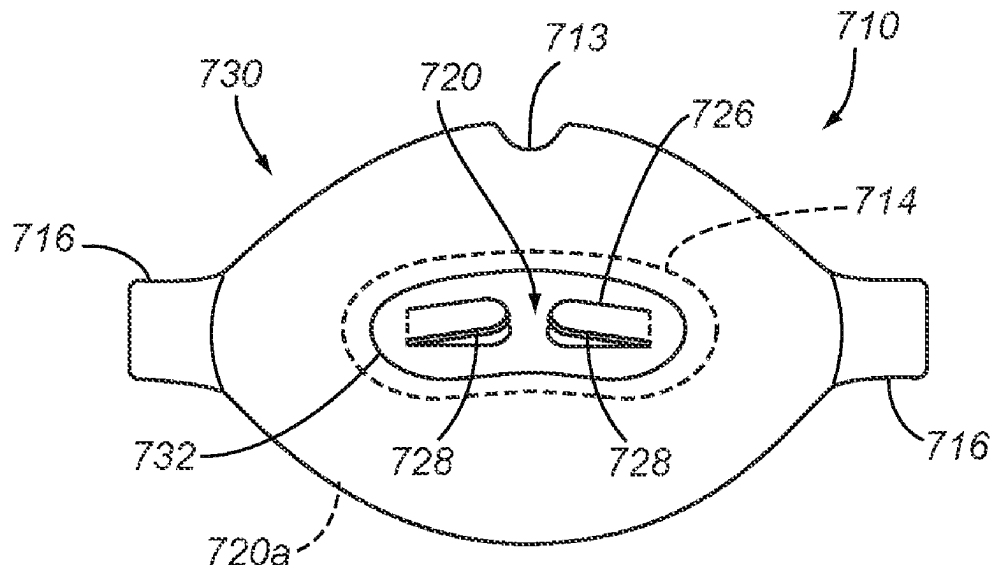
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(57) **ABSTRACT**

Devices and methods are provided for inhibiting mouth breathing and/or improving nasal breathing, e.g., for PAP users or users with other sleeping problems. In an exemplary embodiment, an adhesive device is provided that includes an elongate membrane comprising first and second ends and side edges extending between the first and second ends, a first surface comprising an adhesive layer. The adhesive layer may be a low tack adhesive that is adherent to skin, yet is also easily removable without leaving significant residue. Optionally, the device may include one or more valves to provide one-way flow across the membrane, one or more weakened regions in the membrane, and/or one or more tabs to facilitate removal of the membrane from over a user's mouth.

20 Claims, 8 Drawing Sheets



Related U.S. Application Data

- continuation-in-part of application No. 14/200,010, filed on Mar. 6, 2014, now Pat. No. 9,849,260.
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A61M 39/24 (2006.01)
- (52) **U.S. Cl.**
CPC *A61M 16/0688* (2014.02); *A61M 39/24* (2013.01); *A61M 2039/2426* (2013.01); *A61M 2039/2433* (2013.01)

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CPC A61M 2039/2433; A61F 5/37; A61F 5/56; A63B 23/18; A41D 13/11; A41D 13/1161
See application file for complete search history.

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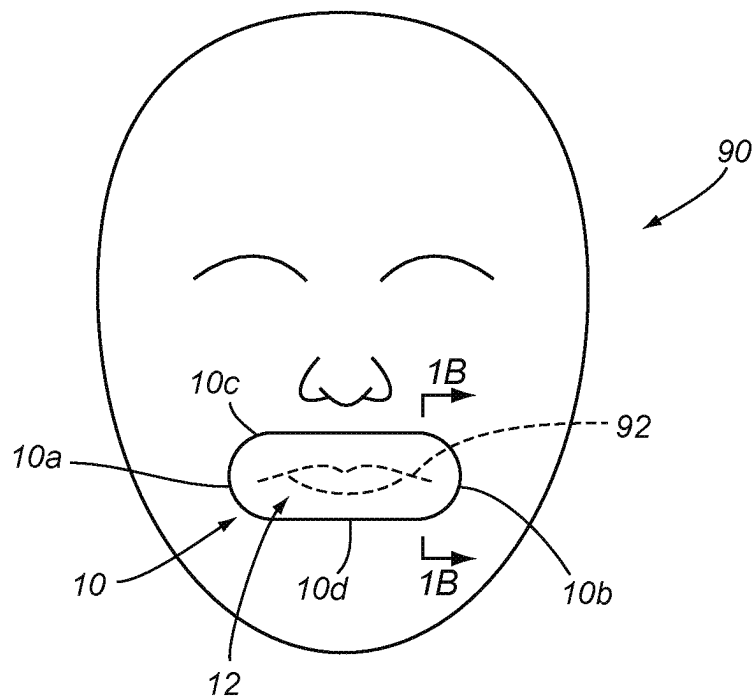


FIG. 1A

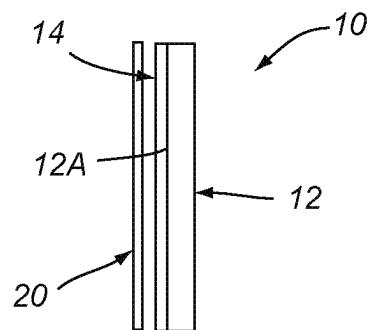


FIG. 1B

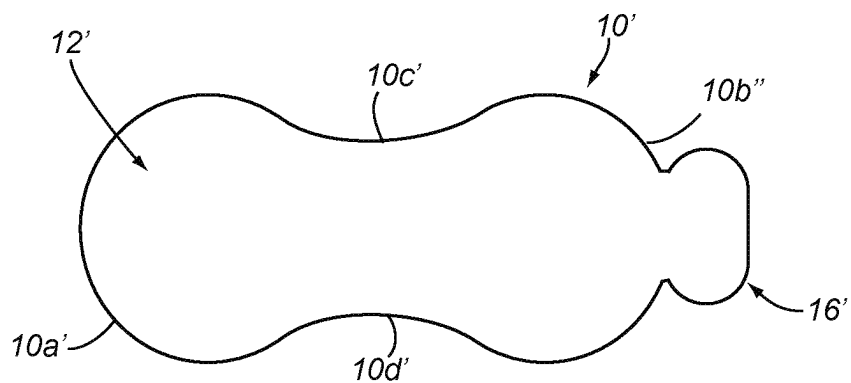


FIG. 2A

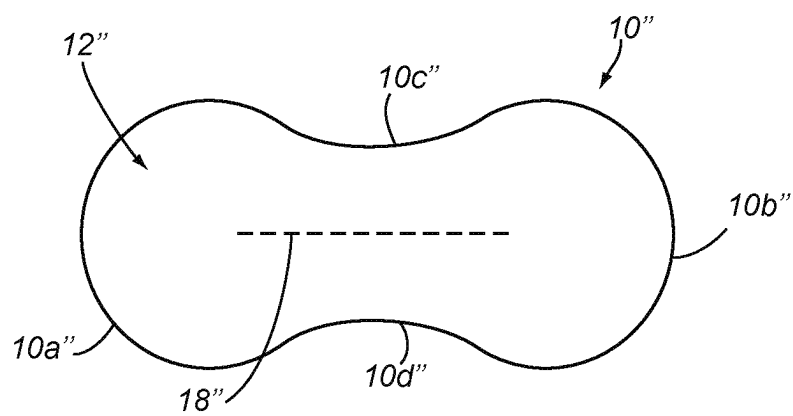


FIG. 2B

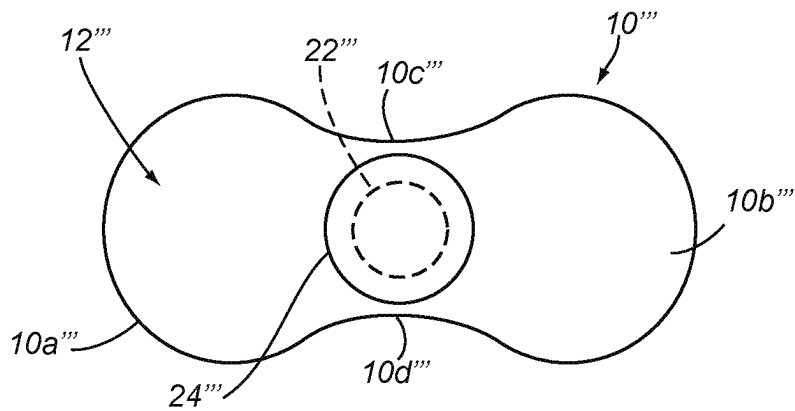
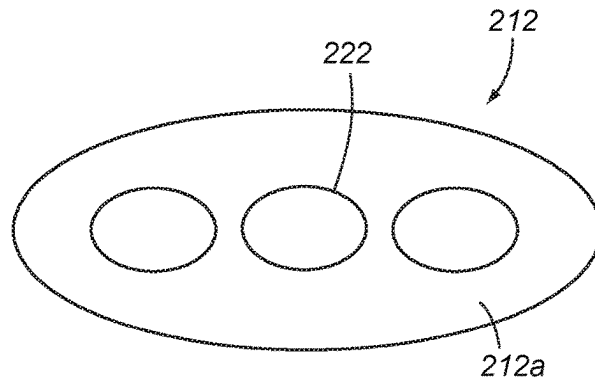
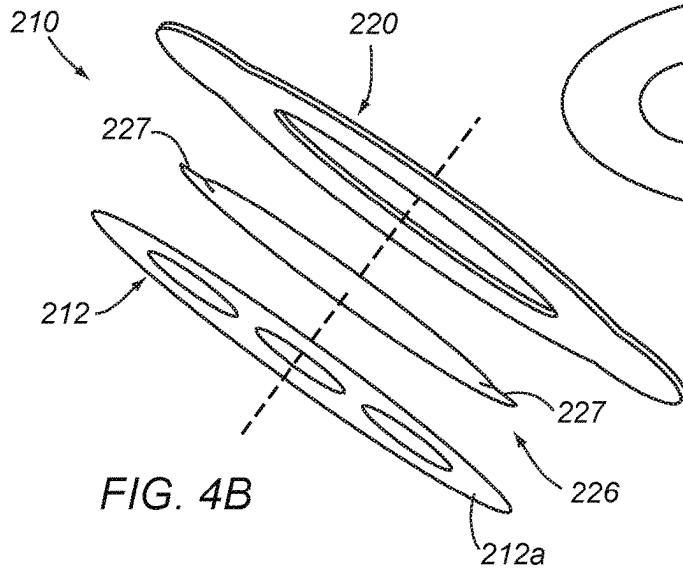
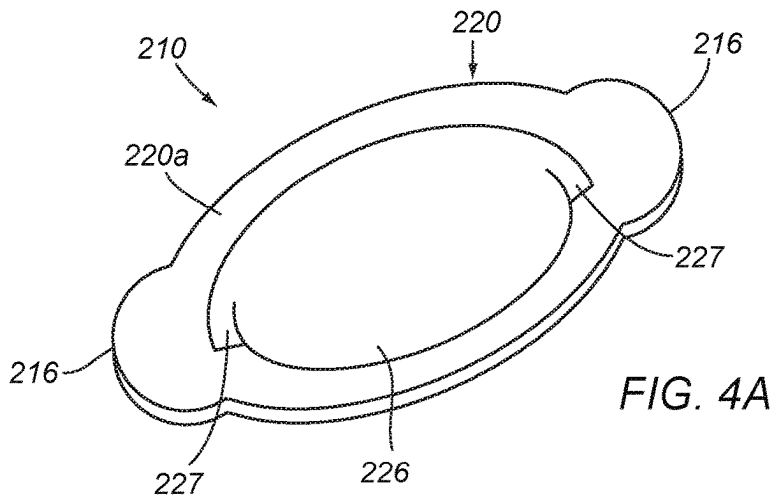
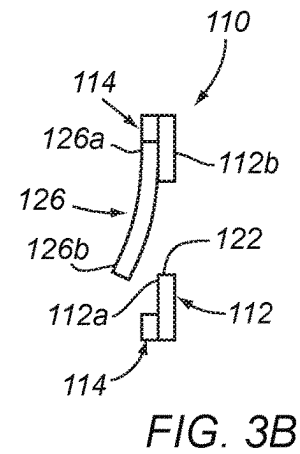
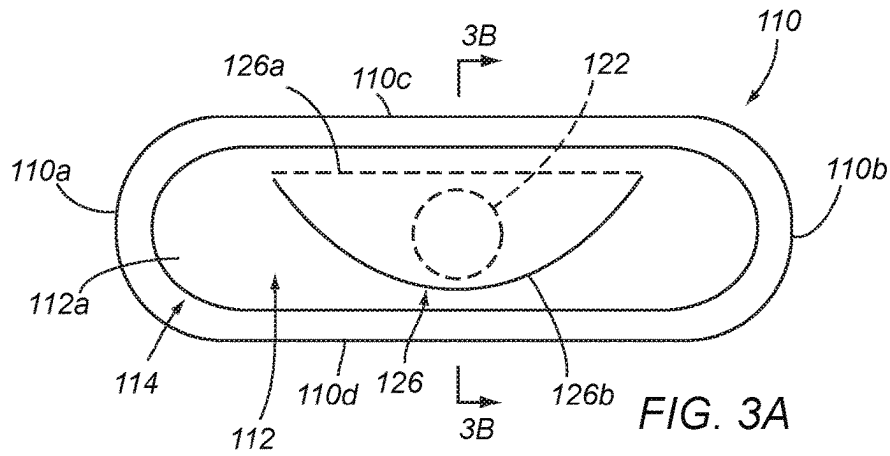


FIG. 2C



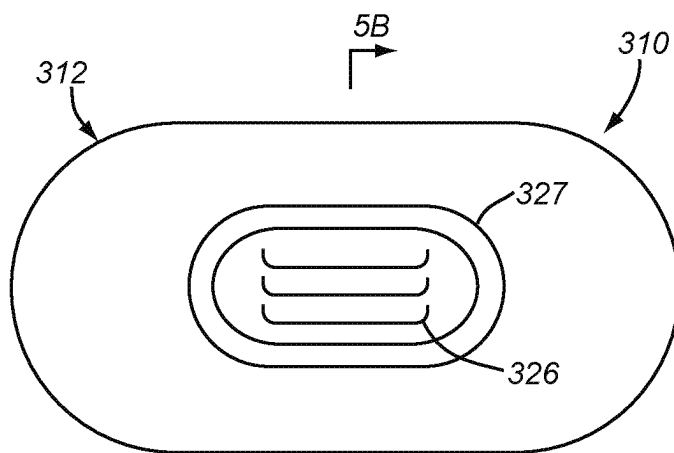


FIG. 5A

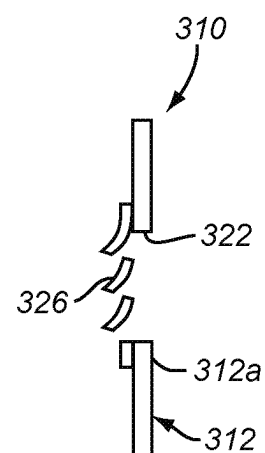


FIG. 5B

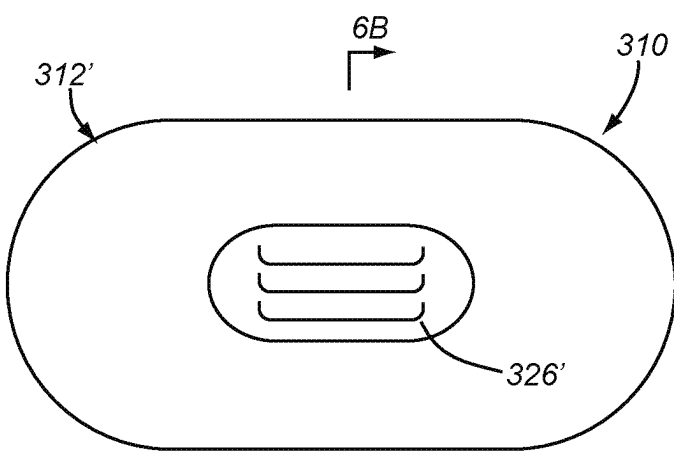


FIG. 6A

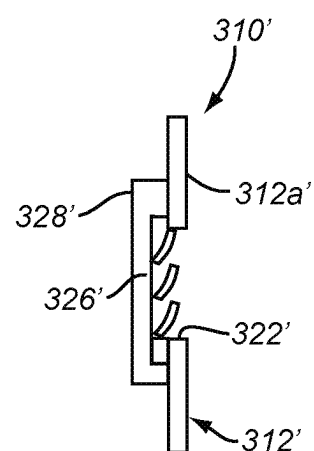
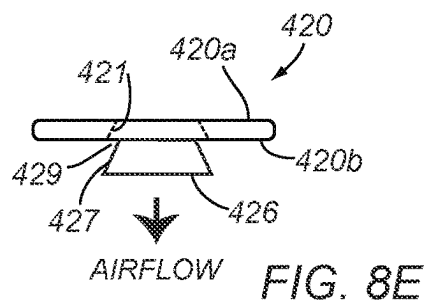
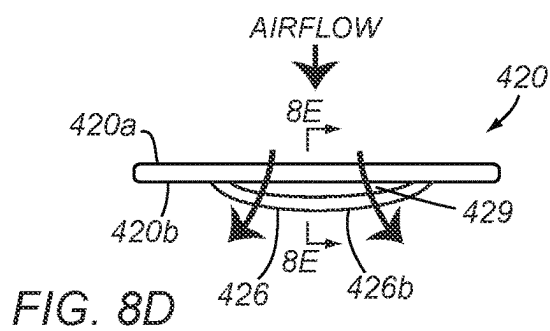
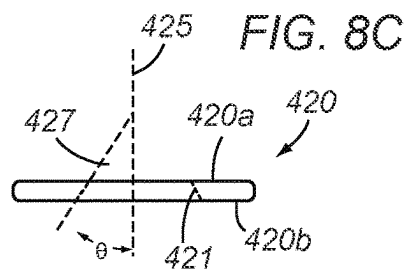
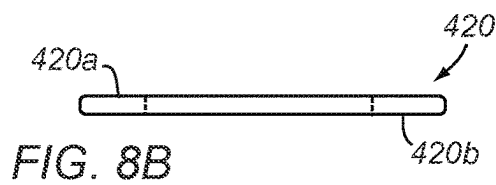
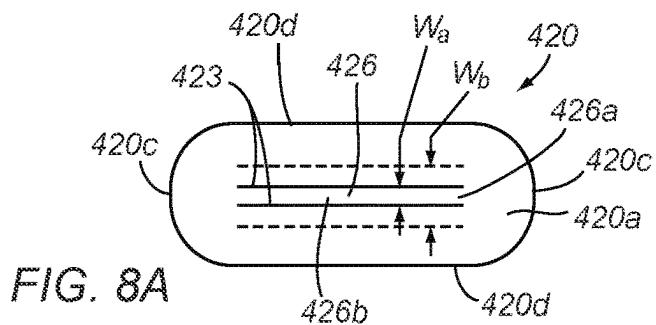
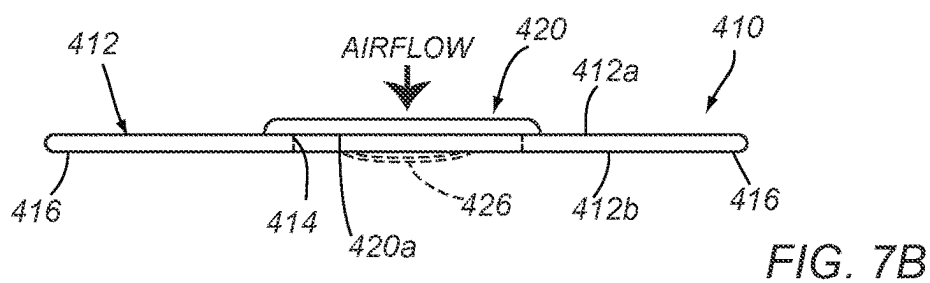
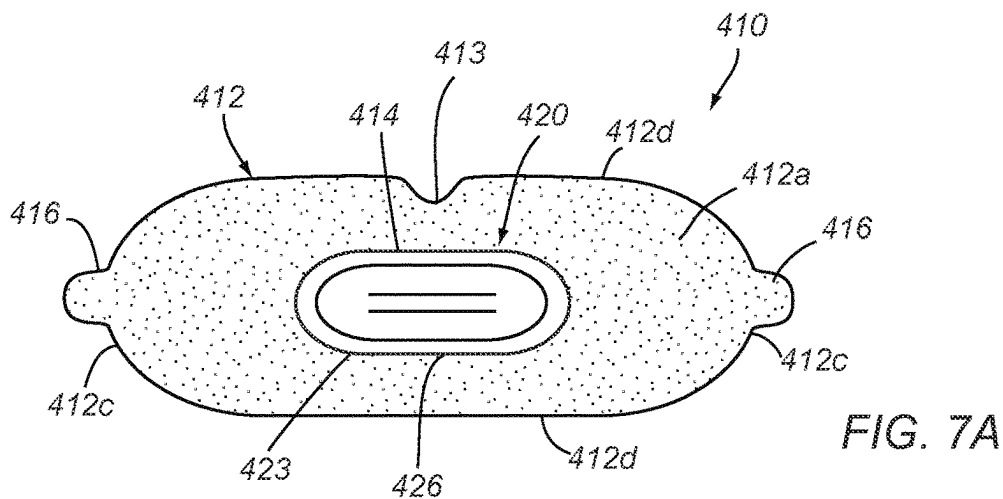


FIG. 6B



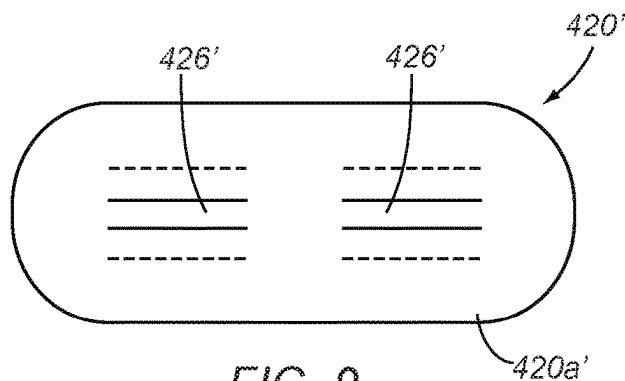


FIG. 9

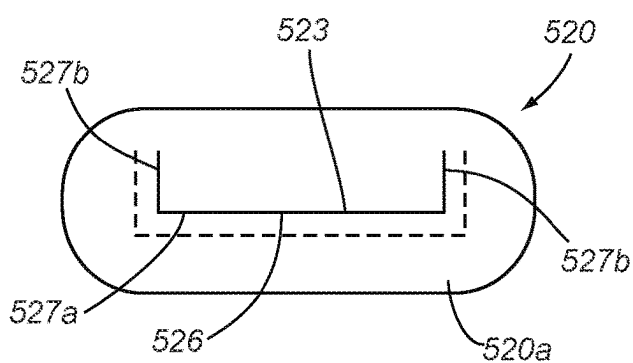


FIG. 10A

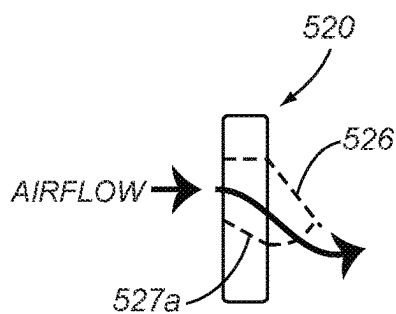


FIG. 10B

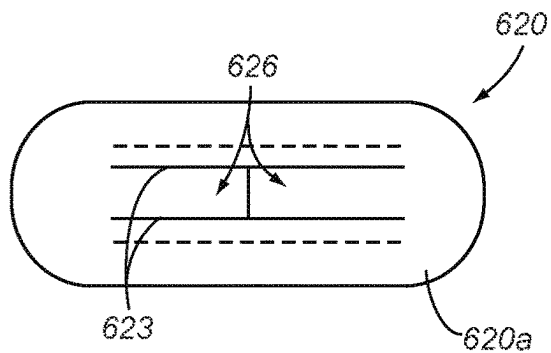


FIG. 11A

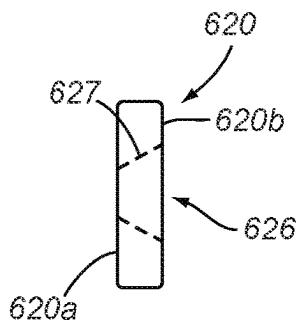


FIG. 11C

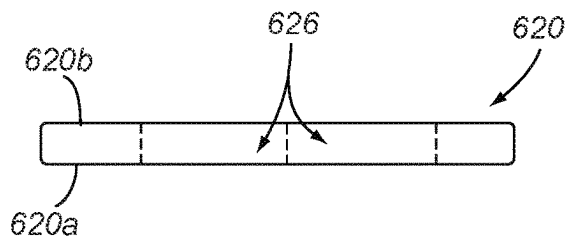


FIG. 11B

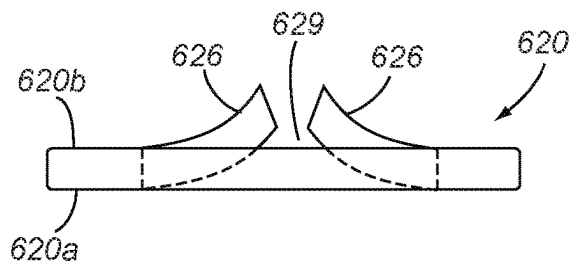


FIG. 11D

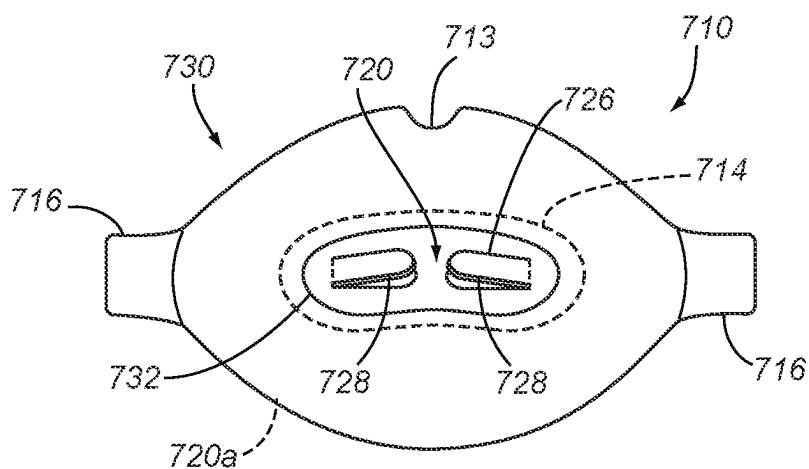


FIG. 12

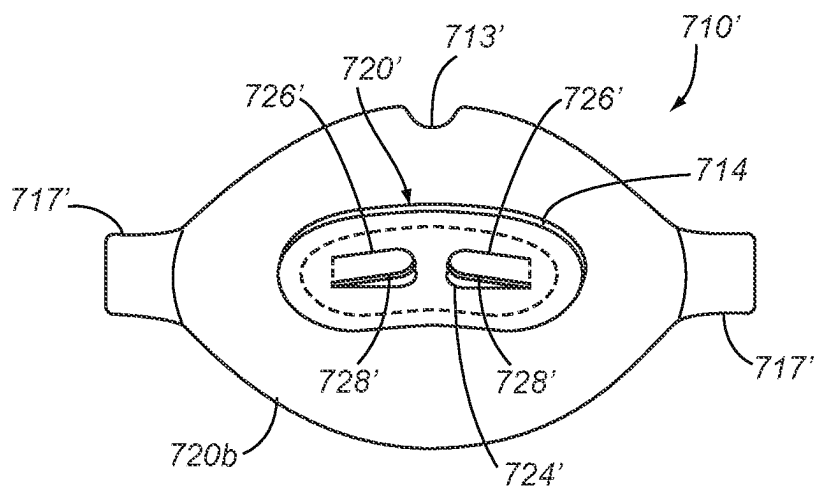


FIG. 13

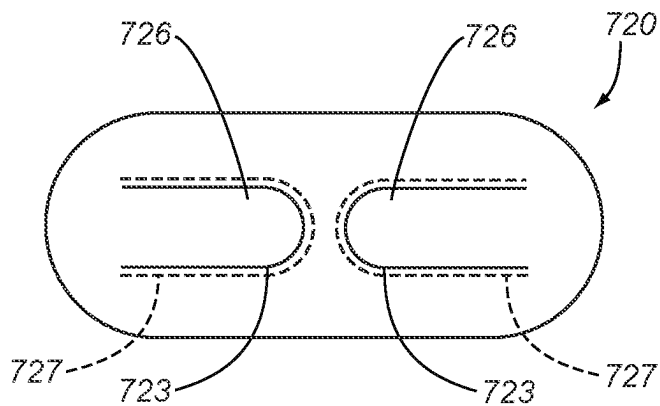


FIG. 14

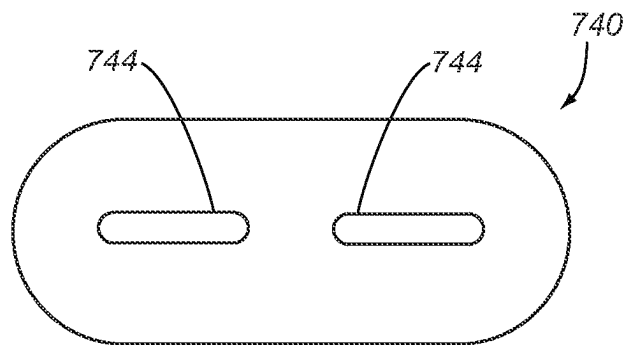


FIG. 15

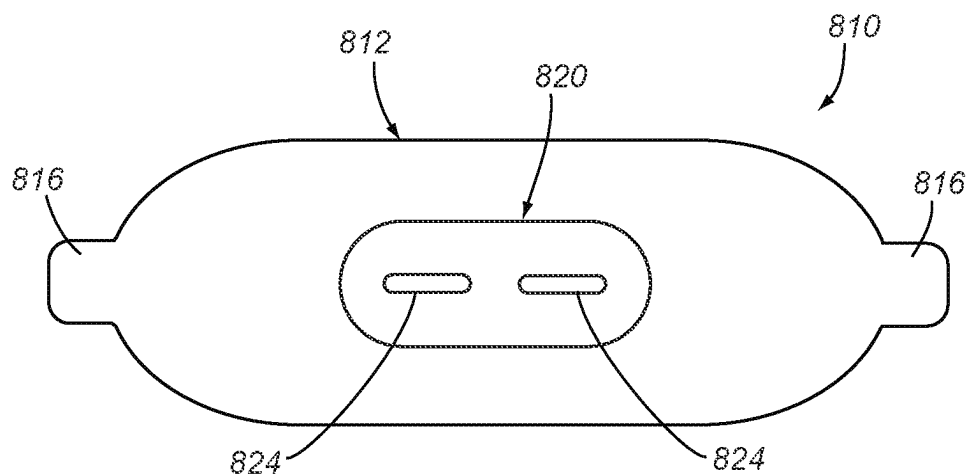


FIG. 16A

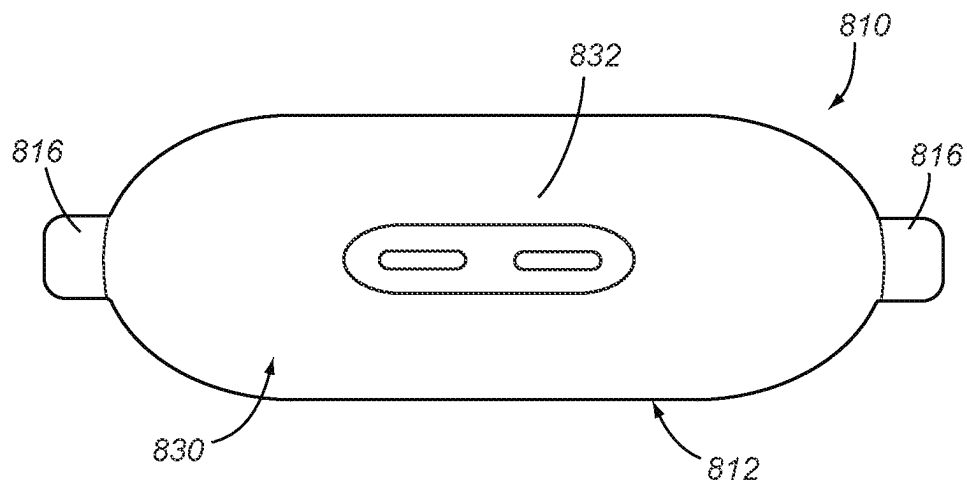


FIG. 16B

DEVICES AND METHODS FOR ENCOURAGING NASAL BREATHING

This application is a continuation of application Ser. No. 15/224,510, filed Jul. 29, 2016, which claims benefit of U.S. provisional application Ser. No. 62/198,132, filed Jul. 29, 2015, and 62/351,142, filed Jun. 16, 2016, and which is a continuation-in-part of U.S. application Ser. No. 14/200,010, filed Mar. 6, 2014, now U.S. Pat. No. 9,849,260, which claims benefit of U.S. provisional application Ser. Nos. 61/773,711, filed Mar. 6, 2013, and 61/816,380, filed Apr. 26, 2013, the entire disclosures of which are expressly incorporated by reference herein.

FIELD OF THE INVENTION

The present invention relates generally to devices and methods for improving breathing and/or sleep, e.g., to adhesive devices for application over a user's mouth to hold the lips, jaw, and/or mouth closed to facilitate nasal breathing, e.g., during use of a positive airway pressure (PAP) or oral pressure therapy (OPT) device and/or to reduce snoring or other breathing problems while sleeping, and/or to improve sleep-related dry mouth and dry-mouth-related oral and dental disease.

BACKGROUND

For a number of reasons and in a variety of conditions, mouth breathing causes or exacerbates snoring, sleep apnea, and/or dry mouth. For example, during use of nasal PAP devices for sleep apnea, positive pressure may be applied to the nose via a nasal mask. However, an open mouth allows the applied positive pressure to simply escape through the mouth rather than open the airway, thus negating the therapeutic value of PAP. Further, even outside of PAP use, breathing through the mouth or having an open mouth may increase snoring, apneas, dry mouth, and the like compared to natural nasal breathing.

A range of products have been proposed to address these problems. For example, PAP full face masks may be used to cover both the nose and the mouth of a user. However, such masks may involve challenges such as sealing and comfort. In addition, chin straps may be used in an attempt to hold up the chin up and keep the mouth closed.

In addition, during normal sleep, jaw opening and/or retraction may encourage collapse of the posterior pharynx and increase airway obstruction. Mouth breathing during sleep also dries the airway. Both of these conditions may encourage snoring. Dry mouth can also cause or exacerbate mouth and gum tissue deterioration and disease and/or tooth decay and disease.

Accordingly, devices and methods for improving breathing, sleep, and/or mouth dryness would be useful.

SUMMARY

The present invention is directed generally to devices and methods for sealing the lips and/or mouth, maintaining jaw position, encouraging natural nasal breathing and/or otherwise improving breathing and/or sleep. More particularly, the present invention is directed to adhesive devices for application over a user's mouth to hold the mouth closed to facilitate normal nasal breathing, e.g., during use of a PAP device, to encourage normal nasal breathing, reduce snoring

or other breathing problems while sleeping, and/or to reduce sleep-related dry mouth, and to methods for using such devices.

Lips have the natural ability to seal and are quite elastic. Even when the jaw drops and/or opens, the lips may stay closed if held closed. However, when the jaw is open and the individual is asleep, the lips tend to relax and the mouth may open, compromising use of PAP devices and/or increasing snoring and/or other breathing problems, e.g., due to mouth breathing. The force required to keep the lips closed even with the jaw open is minimal, especially compared to the force required to keep the chin up and jaw closed in the typical snorer or sleep apnea sufferer. Thus, a novel and helpful device may generally keep the lips closed while providing a number of related features to ensure safety and/or comfort.

In accordance with one embodiment, a device is provided for improving breathing and/or inhibiting mouth breathing that includes a mouth cover comprising an annular member defining a central open region therethrough and including a first surface comprising adhesive thereon for detachably attaching the mouth cover to skin of a user surrounding the user's mouth; and a lip closing element comprising an elastic membrane attached to a second surface of the mouth cover opposite the first surface and extending across the central open region and including one or more openings therethrough.

Optionally, the device may include one or more valves, e.g., a one-way valve attached to the mouth cover, for selectively sealing the one or more openings. For example, when a positive pressure within a user's mouth is generated between the first and second surfaces, the one or more valves may seal the one or more openings, while, when a user inhales to generate a negative pressure between the first and second surfaces, the one or more valves may open the one or more openings.

In accordance with another embodiment, a device for inhibiting mouth breathing is provided that includes an elongate membrane comprising first and second rounded ends and side edges extending between the first and second ends, the elongate member including a first surface comprising an adhesive layer, the adhesive layer comprising a low tack adhesive that is adherent to skin, yet is also easily removable without leaving significant residue. Optionally, the device may include one or more weakened regions in the membrane, the one or more weakened regions configured to separate and allow breathing through the user's mouth when the user forcibly separates the user's lips and/or inhales or exhales above a predetermined pressure threshold. In addition or alternatively, the device may include one or more tabs extending from the membrane, each tab having non-adhesive surfaces and configured to facilitate removal of the membrane from over a user's mouth.

In accordance with yet another embodiment, a device is provided for inhibiting mouth breathing that includes a membrane comprising first and second ends and side edges extending between the first and second ends; an adhesive layer on a first surface of the membrane adherent to skin to allow the membrane to be detachably attached over a user's mouth; and a tab extending from the membrane comprising non-adhesive surfaces and configured to facilitate removal of the membrane from over a user's mouth.

In accordance with another embodiment, a method is provided for improving breathing of a PAP user that includes placing a nasal mask for a PAP device over a user's nose;

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applying a membrane over the user's mouth to prevent breathing through the user's mouth; and activating the PAP device.

In accordance with still another embodiment, a method is provided for improving nasal breathing that includes applying a membrane over a user's mouth with the user's lips pressed together; and sleeping with the membrane over the user's mouth to inhibit breathing through the user's mouth.

In accordance with another embodiment, a device is provided for improving breathing that includes a mouth cover comprising a base membrane including first and second surfaces opposite one another, a central open region therethrough, and an adhesive on at least a portion of the first surface for detachably attaching the mouth cover to skin of a user surrounding the user's mouth; and a valve membrane attached to one of the first and second surfaces of the base member such that the valve membrane extends across the central open region, the valve membrane including one or more valve elements adjacent the central open region, the one or more valve elements including beveled edges adjacent corresponding beveled edges of the valve membrane such that the one or more valve elements are movable away from the valve membrane in a first direction to create one or more openings through the valve membrane to allow airflow in the first direction through the central open region, the beveled edges seating the one or more valve elements in the one or more openings to prevent the one or more valve elements from moving in a second direction opposite the first direction and preventing airflow in the second direction.

In accordance with still another embodiment, a device is provided for improving breathing that includes a base membrane including first and second surfaces opposite one another, a central open region therethrough, and an adhesive on at least a portion of the first surface for detachably attaching the mouth cover to skin of a user surrounding the user's mouth; and a valve membrane attached to one of the first and second surfaces of the base member such that the valve membrane extends across the central open region, the valve membrane including a pair of longitudinal slits formed therethrough that are spaced apart from one another to define a valve element therebetween, the valve element comprising opposite ends that are substantially fixed relative to adjacent material of the valve membrane and an intermediate region that is free to move relative to the adjacent material, wherein the slits are formed such that the valve element includes beveled edges adjacent corresponding beveled edges of the valve membrane that extend at an acute angle relative to an axis normal to the first surface of the base membrane.

In accordance with yet another embodiment, a device is provided for improving breathing that includes a base membrane including first and second surfaces opposite one another, a central open region therethrough, and an adhesive on at least a portion of the first surface for detachably attaching the mouth cover to skin of a user surrounding the user's mouth; and a valve membrane attached to one of the first and second surfaces of the base member such that the valve membrane extends across the central open region, the valve membrane including one or more valve elements adjacent the central open region, each valve element comprising a flap including a first end substantially fixed relative to adjacent material of the valve membrane, a second free end movable out of a plane defined by the valve membrane in a first direction to create an opening to allow airflow through the valve membrane in the first direction, the flap further comprising one or more beveled side edges that prevent the flap from moving in a second direction opposite the first direction to prevent airflow in the second direction.

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In accordance with another embodiment, a method is provided for improving breathing that includes applying a membrane over a user's mouth, the membrane comprising one or more valve elements including beveled edges such that the valve elements open freely in a first direction when the user applies a first pressure to the membrane through the user's mouth to allow airflow through the membrane in the first direction, and wherein the beveled edges seat against adjacent beveled edges of the membrane when the user applies a second pressure opposite to the first pressure to the membrane through the user's mouth to prevent substantial airflow in the second direction.

Other aspects and features of the present invention will become apparent from consideration of the following description taken in conjunction with the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

The invention is best understood from the following detailed description when read in conjunction with the accompanying drawings. It will be appreciated that the exemplary devices shown in the drawings are not necessarily drawn to scale, with emphasis instead being placed on illustrating the various aspects and features of the illustrated embodiments.

FIG. 1A is a front view of an exemplary embodiment of an adhesive device applied over a user's mouth.

FIG. 1B is a cross-sectional side view of the adhesive device of FIG. 1A taken along line 1B-1B.

FIGS. 2A-2C are front views of alternative embodiments of an adhesive device for application over a user's mouth.

FIG. 3A is a front view of another embodiment of an adhesive device.

FIG. 3B is a cross-sectional view of the adhesive device of FIG. 3A taken along line 3B-3B.

FIGS. 4A and 4B are perspective and exploded views, respectively, of another embodiment of an adhesive device.

FIG. 4C is a front view of a lip closing element that may be included in the adhesive device of FIGS. 4A and 4B.

FIGS. 5A and 6A are front views of additional embodiments of adhesive devices.

FIGS. 5B and 6B are cross-sectional views of the adhesive devices of FIGS. 5A and 6A taken along lines 5B-5B and 6B-6B, respectively.

FIGS. 7A and 7B are front and top views, respectively, of another embodiment of a breathing device including an adhesive base member and a valve membrane attached to the base member including a valve element.

FIG. 8A is a front view of an exemplary embodiment of a valve membrane including a valve element that may be included in the breathing device of FIGS. 7A and 7B.

FIGS. 8B and 8C are side views of the valve membrane of FIG. 8A.

FIGS. 8D and 8E are top and side views, respectively, of the valve membrane of FIG. 8A, showing the valve element of the valve membrane stretched to an open configuration to allow air to pass through the valve membrane.

FIG. 9 is a front view of another embodiment of a valve membrane including a pair of valve elements that may be included in the breathing device of FIGS. 7A and 7B.

FIGS. 10A and 10B are front and side views of yet another embodiment of a valve membrane that may be included in the breathing device of FIGS. 7A and 7B.

FIG. 11A is a front view of yet another embodiment of a valve membrane including a pair of valve elements that may be included in the breathing device of FIGS. 7A and 7B.

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FIGS. 11B and 11C are side views of the valve membrane of FIG. 11A.

FIG. 11D is a side of the valve membrane of FIG. 11A with valve elements open.

FIG. 12 is a perspective view of a breathing device configured to permit inhalation and prevent exhalation, and including a valve membrane with beveled valve flaps.

FIG. 13 is a perspective view of a breathing device configured to permit exhalation and prevent inhalation, and including a valve membrane with beveled valve flaps.

FIG. 14 is a detail showing an exemplary embodiment of a valve membrane that may be included in the breathing devices of FIG. 12 or 13.

FIG. 15 is a front view of an exemplary embodiment of a backing member that may be included on a breathing device opposite valve flaps of a valve membrane.

FIGS. 16A and 16B are front and back views of a breathing device including a valve membrane with open valve passages.

DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

Turning to the drawings, FIGS. 1A and 1B show an exemplary embodiment of an adhesive device 10 that generally includes a flexible membrane 12 sized for application over a user's mouth 92 and an adhesive layer 14. Optionally, the adhesive device 10 may also include a removable cover sheet 20, e.g., as shown in FIG. 1B, which may be provided over the adhesive layer 14 to prevent exposure before use.

The membrane 12 may be formed from nonporous material, e.g., to substantially seal the user's mouth to prevent air from passing therethrough. Alternatively, the membrane 12 may simply be configured to maintain the user's lips together and/or keep the user's jaw in a closed position, e.g., such that the lips themselves prevent substantial airflow or breathing through the user's mouth.

In an exemplary embodiment, the membrane 12 may be formed from a non-woven polymeric material that is substantially nonporous, e.g., to prevent airflow through the membrane during normal breathing. Alternatively, the membrane 12 may be formed from a non-woven yet porous material that allows airflow therethrough above a predetermined pressure threshold.

In a further alternative, the membrane 12 may be formed from a woven material or may include filaments or other support materials (not shown) embedded or otherwise attached to a base material. The material may be nonporous or may be porous, e.g., to allow airflow in one or both directions through the membrane. For example, the material may be a mesh that allows limited airflow through the membrane 12 while holding the lips preferentially together.

In addition, the membrane 12 may be formed from compliant material, e.g., for comfort and/or to conform to the shape of the user's mouth and/or surrounding facial features. For example, the membrane 12 may be formed from substantially inelastic material that has sufficient flexibility to conform to the shape of the user's mouth yet support the user's lips and/or jaw. Alternatively, the membrane 12 may be formed from elastic material that is biased to a relaxed profile yet may be resiliently expanded to a stretched profile, e.g., if desired for comfort and/or conformance to the patient's mouth and/or allow some jaw movement.

Further alternatively or in addition, the membrane 12 may be formed from a breathable material that allows passage of moisture away from the skin.

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Further alternatively or in addition, the membrane 12 may have a biased compliance or elasticity. For example, the membrane 12 may be relatively more stretchable or elastic in the horizontal direction between the ends 10a, 10b of the adhesive device 10, e.g., substantially parallel to the part line of the lips, and relatively less stretchable or elastic in the vertical direction between the side edges 10c, 10d of the adhesive device 10, e.g., at a right angle to the part line of the lips.

As shown in FIG. 1B, the adhesive layer 14 may be provided substantially continuously on a first or inner surface 12a of the membrane 12. Alternatively, the adhesive layer 14 may be provided discontinuously on the inner surface 12a. For example, the adhesive layer 14 may be provided around an outer periphery of the inner surface 12a, e.g., in an annular layer such that the adhesive layer 14 may contact the user's skin surrounding the lips while a central region has no adhesive material, e.g., to prevent the adhesive from contacting the user's lips. In further alternatives, other configurations may be provided for the adhesive layer 14, e.g., only on end regions 10a, 10b of the adhesive device 10 and not on a central region, and the like. For example, the adhesive layer 14 may be disposed such that it covers the inner surface 12a approximately one third of the width of the lips on either side and does not cover the middle one third of the width of the lips (e.g., about one third the length of the adhesive device 10 from each end region 10a, 10b and not one third of the length corresponding to the central region).

In an exemplary embodiment, the adhesive layer 14 may be formed from a low tack adhesive that may be adherent to skin, yet is also easily removable without leaving significant residue. The adhesive may be selected to provide low irritation and/or be hypoallergenic to prevent irritation or adverse reaction by the user's skin. In addition or alternatively, the adhesive layer 14 may be able to maintain a substantial bond with the user's skin even if it becomes damp. Alternatively, the adhesive layer 14 may be designed to separate from skin at a predetermined pressure, e.g. in the range that may be generated by forced exhalation. In exemplary embodiments, the adhesive layer 14 may comprise one or more of a silicone adhesive, acrylic adhesive, or other types of adhesive.

As shown in FIG. 1A, the adhesive device 10 may have an elongated shape configured to cover a user's mouth, e.g., having rounded first and second ends 10a, 10b and substantially straight side edges 10c, 10d extending between the ends 10a, 10b. In exemplary embodiments, the adhesive device 10 may have a length between the ends 10a, 10b of between about five and ten centimeters (5-10 cm), and a width between the side edges 10c of between about one and five centimeters (1-5 cm).

Alternatively, the adhesive device may have other shapes. For example, as shown in FIG. 2A, the adhesive device 10' may include curvilinear side edges 10c', 10d' extending between the ends 10a', 10b', having a relatively narrower width at a central region of the adhesive device 10' and a relatively wider width adjacent the ends 10a', 10b'. In this alternative, the adhesive layer (not shown) may be provided on the inner surface of the wider end regions and not the narrower central region.

In addition or alternatively, any of the adhesive devices herein may include a tab, handle, and/or other feature(s) to facilitate removal of the adhesive device 10' after use. For example, as shown in FIG. 2A, a tab 16' may be provided that extends from one end 10b' of an adhesive device 10'. The inner surface of the tab 16' may not have an adhesive layer, e.g., such that the tab 16' does not adhere to the user's

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skin while the adjacent end **10b'** and other regions of the adhesive device **10'** adhere to the user's skin surrounding the user's mouth to provide a desired seal and/or support. Thus, without adhesive on the surfaces of the tab **16'**, the tab **16'** may be easily grasped and pulled to remove the adhesive device **10'**.

In addition or alternatively, any of the adhesive devices herein may include one or more features for selectively allowing passage of air under predetermined conditions. For example, as shown in FIG. 2B, an adhesive device **10"** may be provided that includes one or more weakened regions **18"** in the membrane **12"**. As shown, the weakened regions **18"** may include a plurality of indentations or thinned regions that extend partially through the membrane **12"**, i.e., partially between the inner and outer surfaces. Alternatively, the weakened regions **18"** may include perforations extending entirely through the membrane **12"**. In an exemplary embodiment, the weakened regions **18"** may be aligned spaced apart from one another in a line extending at least partially between the ends **10a**, **10b** of the adhesive device **10**, as shown in FIG. 2B. Alternatively, a plurality of separate weakened regions may be provided (not shown), e.g., that are spaced apart from one another between the ends of the adhesive device such that each region may selectively tear, separate, or otherwise open to provide individual openings through the adhesive device.

For example, the membrane **12"** of the adhesive device **10"** may remain intact with the weakened regions **18"** sealed until a predetermined pressure threshold is exceeded, whereupon the weakened regions **18"** may tear, separate, or otherwise open to allow airflow. For example, the weakened regions **18"** may be configured to separate when a deep inspiration above a predetermined pressure threshold is generated by the user, or when a forced exhalation occurs that is above the predetermined pressure threshold (e.g., greater than the positive pressures typically encountered using PAP devices). In addition or alternatively, the weakened regions **18"** may be configured to open when a predetermined force threshold is applied between the side edges **10c**, **10d** of the adhesive device **10**, e.g., if the user tries to forcibly separate their lips or otherwise open their mouth. In a further alternative, the adhesive layer itself may have predetermined adhesive strength such that the adhesive may separate from the user's skin under such predetermined conditions.

Turning to FIG. 2C, another embodiment of an adhesive device **10'''** is shown that includes a membrane **12'''** and an adhesive layer (not shown) similar to other embodiments, except that the membrane **12'''** includes one or more openings (one opening **22'''** shown) therethrough, and a detachable patch **24'''** over the opening **22'''** (or over each opening if multiple openings are provided). For example, the material of the membrane **12'''** may be sufficiently strong to resist tearing or other separation during normal use. The patch **24'''** may be formed from a film or other material, e.g., similar to the membrane **12'''** or a different material that may be attached to the membrane **12'''** over the opening **22'''**.

The patch **24'''** and/or the membrane **12'''** may include adhesive to detachably attach the patch **24'''** to the outer surface of the membrane **12'''** over the opening **22'''**. For example, a continuous layer of adhesive (not shown) may be provided around the perimeter of the patch **24'''** and/or around the opening **22'''** such that the patch **24'''** may be attached to the membrane **12'''** to substantially seal the opening **22'''** from airflow. Alternatively, if desired, the layer of adhesive may be discontinuous (also not shown), e.g., to allow limited flow of air around the patch **24'''** into and/or

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out of the opening **22'''**. In addition, the adhesive may be selected such that the adhesive may fail if a predetermined pressure threshold occurs across the opening **22'''**, e.g., to cause the patch **24'''** to at least partially separate from the membrane **12'''** and allow airflow through the opening **22'''**.

Turning to FIGS. 3A and 3B, another exemplary embodiment of an adhesive device **110** is shown that includes a one-way valve or flow limiting/prevention device that selectively allows airflow through the adhesive device under predetermined conditions. For example, a one-way valve **126** may be incorporated into the membrane **112** that allows air to flow freely into the user's mouth (i.e., from the outer surface **112b** through the inner surface **112a**), or that opens only when a predetermined pressure threshold is exceeded to allow airflow. Alternatively, the one-way valve **126** may allow airflow from the user's mouth (i.e., from the inner surface **112a** through the outer surface **112b**) freely or when a predetermined pressure threshold is exceeded. In a further alternative, the one-way valve **126** may allow limited airflow into or out of the user's mouth below a predetermined pressure threshold and increased airflow into or out of the user's mouth above the predetermined pressure threshold.

In the embodiment shown, the adhesive device **110** includes a membrane **112** including an opening **122** therethrough, e.g., in a central region thereof, and an adhesive layer **114** around a periphery of the inner surface **112a** of the membrane **112**. In addition, a flap **126** is integrally formed with or attached to the membrane **112**, e.g., such that the flap **126** covers the opening **122**.

In one embodiment, the flap **126** may be a separate flexible membrane or film, e.g., formed from similar or compatible materials with the membrane **112**, with one edge of the flap **126a** attached to the inner surface **112a** of the membrane **112** and an opposite, e.g., curved, edge **126b** that remains free. Alternatively, the curved edges **126b** may be detachably attached to the inner surface **112a** of the membrane **112**, e.g., using an adhesive that separates at a predetermined pressure threshold, similar to the patch **24'''** of the adhesive device **10'''** shown in FIG. 2C.

In another embodiment, the membrane **112** may include multiple layers (not shown), e.g., a first or outer layer that includes the opening **122** and a second or inner layer that includes the flap **126**, e.g., by simply cutting or otherwise separating the free edge **126b** from the rest of the second layer. The layers may be attached together, e.g., substantially permanently, by one or more of bonding with adhesive, fusing, sonic welding, and the like, except that the area of the flap **126** such that the flap **126** is free to move away from and towards the opening **122**.

Thus, during use, the adhesive device **110** may be applied to a user's face, e.g., around the user's mouth similar to the embodiment shown in FIG. 1A such that the flap **126** is disposed adjacent, e.g., against, the user's lips. For example, a cover sheet (not shown) provided over the adhesive layer **114** may be removed, and the adhesive layer **114** may be applied to the user's skin surrounding the mouth, e.g., after the user has pressed their lips together and/or closed his/her mouth/jaw. With the lips pressed together, substantial airflow through the adhesive device **110** may be prevented since the adhesive device **110** may support the lips and/or jaw, e.g., to hold the lips together. Thus, the user may be encouraged and/or limited to breathe nasally rather than through their mouth.

However, if the user's lips separate at all, the flap **126** may prevent airflow out of the user's mouth, e.g., since any positive pressure within the user's mouth applied to the adhesive device **110** may force the flap **126** against the inner

surface of the membrane **112** around the opening **122**. If a negative pressure is created within the user's mouth that is applied to the adhesive device **110**, the pressure may cause the flap **126** to move inwardly away from the opening **122**, thereby allowing airflow through the opening **122**, e.g., into the user's mouth. Alternatively, the flap **126** may be detachably attached to the membrane **112** such that a predetermined pressure threshold is required, e.g., a particular strong inhalation by the user, to cause the flap **126** to at least partially separate and open the opening **122**.

Optionally, any of the adhesive devices herein may be designed to augment sealing of the lips, e.g. by encouraging pursing of the lips or involution of the lips of the user. For example, the user may be instructed to purse their lips before applying the adhesive device, e.g., such that the adhesive device supports and/or holds the lips in the pursed, involuted, or other position, which may enhance sealing and/or undesired airflow through the user's lips.

The adhesive devices herein may be of particular use in conjunction with positive airway pressure, for example, as used to treat sleep apnea. In this setting, mouth leaks can be both bothersome to the patient and decrease efficacy of the treatment. Current methods for preventing mouth breathing are cumbersome and sometimes ineffective. For example, with the adhesive device **110** shown in FIGS. 3A and 3B, positive pressure within the user's mouth, e.g., generated by a PAP nasal device placed over the user's nose, may cause the flap **126** to be pressed against the inner surface **112a** of the membrane **112** to substantially seal the opening **122** and prevent the pressure from the PAP device from leaking out of the users mouth between their lips.

In addition, the adhesive device **110** may hold the lips pressed together and/or hold the jaw closed to prevent mouth breathing. Thus, the user may breathe through their nose, e.g., assisted by the PAP device. In this manner, the adhesive device **110** over the mouth in conjunction with nasal positive pressure may prevent mouth breathing, thereby maintaining therapeutic elevated airway pressure, preventing drying of the mouth, bothersome leaks, and/or other problems often encountered using PAP devices.

In addition, the adhesive devices herein may hold the mouth closed without a PAP device to encourage nasal breathing, thereby reducing snoring, mouth dryness, and/or other problems encountered when the user otherwise breathes through their mouth.

Turning to FIGS. 4A and 4B, another embodiment of an adhesive device **210** is shown that includes a lip closing element **212**, a mouth cover **220** for securing the lip closing element **212** relative to a user's mouth, and a flap **226**. Generally, the mouth cover **220** is an annular member sized and/or shaped to be applied around a user's mouth, e.g., against the skin adjacent the user's lips. Similar to other embodiments herein, the mouth cover **220** includes an adhesive layer (not shown), e.g., on an inner surface **220a** of the mouth cover **220**, for application against the user's skin to secure the adhesive device **210** over the user's mouth.

For example, the mouth cover may be a double-sided adhesive membrane that includes a central or base layer including an inner surface **220a** and an opposite outer surface (not shown), both of which include an adhesive layer. The first or inner adhesive layer on the inner surface **220a** may provide a low tack and/or low residue for securing the adhesive device **210** around a user's mouth, similar to other embodiments herein. The second or outer adhesive layer (not shown) on the outer surface may substantially permanently attach other components of the adhesive device **210** to the mouth cover **220**. The base layer of the mouth

cover **220** may be formed from materials similar to other membranes herein, e.g., providing desired flexibility and/or conformance to a user's face.

Similar to other embodiments herein, the mouth cover **220** may include one or more tabs (two tabs **216** shown on opposite ends of the mouth cover **220**), which do not include any adhesive on the inner or skin contact surface (or other surfaces). Thus, the tab(s) **216** may be easily pulled and/or otherwise separated from the user's skin to facilitate removing the adhesive device **210**, as described elsewhere herein.

With additional reference to FIG. 4C, the lip closing element **212** may be a flexible membrane, e.g., formed from materials similar to any of the other embodiments herein, that may be attached to the outer surface of the mouth cover **220**. For example, an outer periphery of the inner surface **212a** of the lip closing element **212** may be attached to the adhesive layer on the outer surface of the mouth cover **220** to substantially permanently attach them together. In an exemplary embodiment, the lip closing element **212** may be formed from urethane, PEBAX, silicone, or other elastomeric material. Further alternatively, the lip closing element **212** may be formed from a mesh, multiple bands, and/or other substantially air permeable material.

Optionally, as shown, the lip closing element **212** may include one or more openings therethrough, e.g., three openings **222** as shown, to allow the user to breathe, speak, and/or perform other actions when the adhesive device **210** is secured over the user's mouth. For example, the lip closing element **212** may be a membrane formed from elastic material with the openings **222** biased to a relaxed or relatively small area configuration, yet resiliently deflectable to open the openings **222** to an expanded or relatively large area configuration. The support regions between the openings **222** may have sufficient structural integrity and/or elasticity to bias the openings to the small area configuration, yet allow the lip closing element **212** to be stretched elastically to expand the openings. If the lip closing element **212** is formed from relatively low Durometer elastomeric material, the material may need to be thicker or the width of the support elements between the openings wider (e.g., by making the openings **222** smaller or further apart from one another) to provide the same lip closing force as a lip closing element constructed with a relatively higher Durometer material.

Thus, with the adhesive device **210** secured over a user's mouth, the openings **222** may be biased to the small area configuration to minimize airflow through the openings **222**. In the small area configuration, the openings **222** may be substantially circular or may be elliptical, e.g., having the major axis parallel to the length of the adhesive device **210** and the minor axis parallel to the width of the adhesive device **210**. The user may force their lips apart, thereby stretching the lip closing element **212** and/or otherwise expanding the openings **222**, which may facilitate speaking, breathing, removing spittle or other material, and/or taking other actions.

Optionally, the adhesive device **210** may also include one or more flaps or other valve(s) **226** attached to the mouth cover **220** and/or lip closing element **212**. For example, as best seen in FIG. 4B, a flap **226** may be positioned between the mouth cover **220** and lip closing element **212** and secured therebetween, e.g., by tabs **227**. With the tabs **227** secured to the mouth cover **220** and/or lip closing element **212**, the flap **226** may move towards the lip closing element **212**, e.g., to close and/or seal the openings **222**, and/or away from the lip closing element **212**, e.g., into the opening through the mouth cover **220** to allow airflow through the

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openings 222. Further, at least some of the one or more flaps or other valves may be substantially aligned with the part line of the lips. Further if the one or more flaps or other valves are constructed to open inwardly, they may be positioned to open into the space created between the lips when the lips are at least partially opened.

In exemplary embodiments, the flap 226 may be formed from relatively thin, strong, flexible material, such as urethane, PET, PEBAX, and the like. The flap 226 may be formed from inelastic material such that the flap 226 does not stretch but simply moves relative to the lip closing element 212.

For example, if the user separates their lips, the flap 226 may still initially cover the openings 222 in the lip closing element 212. Thus, in this position, positive pressure across the adhesive device 210 (i.e., relatively higher pressure inside the user's mouth than outside) causes the flap 226 to press against the lip closing element 212, sealing the openings. The user may be able to inhale to generate a negative pressure, thereby directing the flap 226 away from the lip closing element 212 and allowing the user to inhale through the openings 222.

If the lip closing element 212 (and mouth cover 220) are sufficiently elastic, the user may be able to open their lips further, i.e., until the flap 226 no longer covers the openings 222. In this further expanded configuration, since the openings 222 are fully open, the user may be able to inhale and exhale, as desired. When the user relaxes their lips and/or mouth, the lip closing element 212 may resiliently close or return towards the relaxed configuration, thereby covering the openings 222 once again with the flap 226.

Optionally, the adhesive device 210 may include one or more additional features, as desired. For example, safety scores (e.g., similar to the weakened regions described elsewhere herein), breakable elements, stretchable elements, and the like (not shown) may be provided on the mouth cover 220 and/or the lip closing element 212. In addition or alternatively, the flap 226 may be tuned to allow for programmed rupture at a given exhalation pressure, with forced opening of the mouth, and the like, e.g., causing the flap material 226 to tear or separate from the tabs 227. The pressure of normal breathing (even with PAP) is much lower than the pressure an individual can readily produce when needed/desired such as in an emergent situation (e.g., if the user needs to vomit, and the like). Thus, the adhesive device 210 (and the other embodiments herein) may be readily fabricated to provide a device that is both safe (e.g., that may break away when needed) and effective (i.e., stay together) under normal use to maintain the user's mouth closed and/or sealed.

During use, the adhesive device 210 may be applied over a user's mouth by attaching the mouth cover 220 to skin surrounding the user's mouth. For example, if a cover sheet (not shown) is provided over the adhesive on the inner surface 220a, the cover sheet may be removed to expose the adhesive, and then the inner surface 220a may be pressed against the skin surrounding the user's mouth. The user may purse their lips and/or otherwise close or position their lips and mouth such that the lip closing element 212 is positioned adjacent the user's lips with the lip closing element 212 in its relaxed or small area configuration.

If the adhesive device 210 includes the flap 226, the flap 226 may be positioned against the user's lips with the lip closing element 212 outside and against the flap 226. Under normal use, the user's lips are held closed to prevent

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breathing in or out through the mouth, e.g., to enhance PAP use, nasal breathing, and/or prevent snoring and/or other undesired mouth breathing.

However, for some users, during the accommodation period when the adhesive device 210 is first applied over the user's mouth, the user may feel a sense of anxiety about the ability to get enough air through their nose. With minimal opening of the lips, air may be easily inspired through the openings 222, as described above, but prevented from escaping, e.g., to encourage good breathing protocol. Further, if needed or desired, exhalation through the mouth may still be readily accomplished if the lips are further opened causing the openings 222 to extend beyond the flap 226, also as explained above. After use, e.g., after sleeping, the adhesive device 210 may be removed, e.g., by pulling one or both tabs 216 to remove the mouth cover 220 from the user's skin.

Turning to FIGS. 5A and 5B, another embodiment of an adhesive device 310 is shown that includes a membrane 312 and an adhesive layer (not shown), generally similar to other embodiments herein. In addition, the adhesive device 310 includes one or more flaps 326 (e.g., three as shown) that may be configured to allow flow in only one direction across the membrane 312, similar to other embodiments herein. In the embodiment shown, the flaps 326 may be formed in a second layer of material, e.g., similar to or compatible with the material of the membrane 312, attached to either side of the membrane 312. For example, the membrane 312 may include a shape similar to any of the embodiments herein with an opening 322 formed therethrough. The flaps 326 may be cut or otherwise formed in a film or membrane having a periphery 327 that is attached to the membrane 312 around the opening 322. The flaps may be configured to move away from and/or towards the opening 322 or otherwise open when a negative pressure is generated across the membrane 312, e.g., when a user inhales, but otherwise close when a positive pressure is generated, e.g., due to PAP use or ordinary exhalation.

In an alternative embodiment shown in FIGS. 6A and 6B, the adhesive device 310' may be formed from three layers of material. For example, the first layer may be a membrane 312' including the opening 322,' and the second layer may include the flaps 326.' In addition, in this embodiment, a third layer 328' is provided that allows the flaps 326' to substantially open in only one direction. For example, the third layer 328' may be a nonporous layer of material, a mesh, a series of separate bands (not shown), and the like, similar to other embodiments herein. The third layer 328' may be substantially inelastic, stretchable in one direction, or substantially elastic. Similar to other embodiments, when a positive pressure is generated across the membrane 312,' the flaps 326' may be at least partially prevented from opening by the third layer 328,' and may thereby create a substantial expiratory seal, yet may open away from the third layer 328' when a negative pressure is generated. The adhesive devices 310, 310' may otherwise be used similar to other embodiments herein.

With reference to FIGS. 6A and 6B, further alternatively, the second layer may be omitted and the flaps 326' prime may be formed in the first layer 312.' In this case, the adhesive layer may be generally omitted from the area of the flaps 326.' The third layer 328,' as previously described, may at least partially prevent the flaps 326' from opening and thereby create a substantial seal when a pressure differential is applied so as to direct the flaps toward the third layer 328,' e.g., due to PAP use or ordinary exhalation.

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Turning to FIGS. 7A and 7B, another embodiment of a breathing device **410** is shown that includes a substantially planar base membrane **412** and a valve membrane **420** attached to the base membrane **412** and configured to limit airflow through the device **410**, which may be constructed similar to other embodiments herein. For example, the base membrane **412** may include one or more layers of base material, e.g., a non-woven material that is substantially nonporous, e.g., to prevent airflow through the membrane during normal breathing. Alternatively, the base membrane **412** may be formed from a non-woven yet porous material that allows airflow therethrough above a predetermined pressure threshold.

In a further alternative, the base membrane **412** may be formed from a woven material or may include filaments or other support materials (not shown) embedded or otherwise attached to a base material. The material may be nonporous or may be porous, e.g., to allow airflow in one or both directions through the membrane. For example, the material may be a mesh that allows limited airflow through the base membrane **412** while holding the lips preferentially together.

In addition, the base membrane **412** may be formed from compliant material, e.g., for comfort and/or to conform to the shape of the user's mouth and/or surrounding facial features. For example, the base membrane **412** may be formed from substantially inelastic material that has sufficient flexibility to conform to the shape of the user's mouth yet support the user's lips and/or jaw. Alternatively, the base membrane **412** may be formed from elastic material that is biased to a relaxed profile yet may be resiliently expanded to a stretched profile, e.g., if desired for comfort and/or conformance to the patient's mouth and/or allow some jaw movement.

As shown in FIG. 7A, the base membrane **412** generally has an elongated shape configured to cover a user's mouth, e.g., having rounded first and second ends **412c** and substantially straight side edges **412d** extending between the ends **412c** such that the base membrane **412** has a height that is smaller than its width. In exemplary embodiments, the base membrane **412** may have a width between the ends **412c** of between about five and fourteen centimeters (5-14 cm), a height between the side edges **412d** between about one and five centimeters (1-5 cm), and a thickness (between the two surfaces **412a**, **412b**) between about 0.375-1.25 mm (0.015-0.050 inch), e.g., about 0.75 mm (0.030 inch).

Alternatively, the base membrane **412** may have other shapes and/or include other features, similar to other embodiments herein. For example, the base membrane **412** may include a notch **413** on one of the side edges, e.g., substantially centered between the ends of the base membrane **412**, e.g., to serve as an alignment guide during application, which may facilitate positioning the device **410** under a user's nose. In addition or alternatively, a tab **416** (e.g., without adhesive) may be provided that extends from one or both ends **412c** (e.g., one at each end, as shown) or, optionally, at other locations around the periphery of the base membrane **412**, e.g., to facilitate removal of the adhesive device **10'** after use, as described elsewhere herein.

The base membrane **412** may be provided in a substantially planar configuration that is substantially flat in a relaxed state yet is sufficiently flexible to be curved and/or otherwise shaped to conform to the user's facial features, e.g., lips, mouth, and/or cheeks. Optionally, the base membrane **412** may have a "curved" substantially planar shape in its relaxed state yet may be directed into other shapes, e.g., to conform to a user's anatomy.

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The base membrane **412** includes a central open region **414**, e.g., extending between the first and second surfaces **412a**, **412b** to provide a flow path across the base membrane **412**, which is limited by the valve membrane **420**, as described further elsewhere herein. In the embodiment shown, the central open region **414** may be an oblong, oval, or other elongated opening aligned along the width of the base membrane **412**, i.e., partially between the ends **412c**. Alternatively, if desired, multiple openings (not shown) may be provided for the central open region **414** instead of a single opening.

The base membrane **412** includes an adhesive material at least partially covering a first surface **412a** of the based membrane **412** (represented by the dot pattern in FIG. 7A), e.g., substantially entirely covering the first surface **412a** other than the one or more removal tabs **416**, similar to other embodiments herein. For example, the adhesive material may be formed from a low tack adhesive that may be adherent to skin, yet is also easily removable without leaving significant residue, i.e., such that the device **410** may be removed without leaving a sticky residue on the user's skin. Optionally, the adhesive material may be designed to separate from skin at a predetermined pressure, e.g., in the range that may be generated by forced exhalation. In exemplary embodiments, the adhesive material may be silicone adhesive, acrylic adhesive, hydrocolloid adhesive, or other types of adhesive applied to the first surface, e.g., by spraying, brushing, and the like, or as a double-sided tape. Alternatively, the adhesive may be provided discontinuously on the first surface **412a**, e.g., around an outer periphery of the first surface **412a** (not shown), e.g., to prevent the adhesive from contacting the user's lips, or otherwise similar to other embodiments herein.

Optionally, a backing material (not shown) may be provided to cover the adhesive before use. For example, a sheet of material, such as paper, nylon, polyester, and the like, may be provided that covers the entire first surface **412a** to protect the adhesive, which may be removed immediately before application to a user's mouth, as described elsewhere herein.

With additional reference to FIGS. 8A-8E, the adhesive device **410** also includes a valve membrane **420** including one or more flaps or other valve elements **426**, which may be configured to allow flow in only one direction across the breathing device **410**, similar to other embodiments herein. The valve membrane **420** may be formed from material that is more flexible and/or elastic than the base membrane **412**, e.g., formed from urethane, PEBAX, silicone, or other elastomeric material, or from polypropylene, polyurethane, polymer foams, paper, or other flexible materials. In exemplary embodiments, the valve membrane **420** may have a thickness between about 0.375-1.25 mm (0.015-0.050 inch), e.g., about 0.75 mm (0.030 inch).

In the embodiment shown, the valve membrane **420** includes first and second surfaces **420a**, **420b** with one of the surfaces **420a** attached to the first surface **412a** of the base membrane **412**, as best seen in FIG. 7B. The valve membrane **420** is sized to cover the central open region **414** of the base membrane **412**, e.g., with an outer periphery of the valve membrane **420** surrounding the opening(s) defining the central open region **414** and the remainder of the first surface **412a** of the base membrane **412** surrounding the valve membrane **420**. For example, the valve membrane **420** may have an oval or oblong shape generally similar to (but smaller than) the base membrane **412**, e.g., including rounded ends **420c** and longitudinal edges **420d** extending between the ends **420c**. The valve membrane **420** may be

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substantially permanently attached to the base membrane 412, e.g., substantially continuously around the outer periphery, e.g. by one or more of bonding with adhesive, fusing, heat welding, sonic welding, and the like.

Generally, the valve element(s) 426 may be formed directly in the material of the valve membrane 420, e.g., by forming one or more slits 423 through the valve membrane 420 (i.e., entirely between the first and second surfaces 420a, 420b) by one or more of mechanical cutting, die cutting, knife cutting, laser cutting, and the like. Alternatively, one or more openings may be formed through the valve membrane 420 and separate valve element(s) may be seated in or adjacent to the opening(s) and attached at one or more locations to the adjacent material of the valve membrane 426, e.g., by one or more of bonding with adhesive, fusing, heat welding, sonic welding and the like such that the valve element(s) fold, hinge, stretch, and/or otherwise open and close, as described further below. In this alternative, the valve elements may be sized to substantially fill the opening. Further, the valve elements may be formed of the same material of the valve membrane or may be formed of material having different desired properties, e.g., elasticity and the like.

In the embodiment shown in FIGS. 8A-8E, a single valve element 426 is provided that is defined by a pair of slits 423 that are spaced apart from one another and extend substantially parallel to one another and the side edges 420d partially between the opposite ends 420c of the valve membrane 420. Thus, opposite ends 426a of the valve element 426 remain substantially fixed relative to the valve membrane 420 while an intermediate region 426b of the valve element 426 is free to move. Alternatively, as shown in FIG. 9, a pair of valve elements 426' (or other number of valve elements, not shown) may be provided in the valve membrane 420, as desired.

Returning to FIGS. 8A-8E, the slits 423 are formed such the valve element 426 defines beveled edges 427 and adjacent material of the valve membrane 420 also define beveled edges 421 to limit movement of the intermediate region 426b. In particular, as best seen in FIG. 8C, the beveled edges 427, 421 extend entirely between the first and second surfaces 420a, 420b of the valve membrane 420 at a non-perpendicular angle relative to the surfaces 420a, 420b. In an exemplary embodiment, the beveled edges 427, 421 define an acute angle θ relative to a normal axis 425 extending perpendicular to the surfaces 420a, 420b, e.g., between about twenty five and sixty five degrees (25-65°). Thus, as shown in FIGS. 8A and 8C, the width w_a of the valve element 426 (i.e., the distance between the slits 423) is narrower at the first surface 420a (the solid line in FIG. 8A) than the width w_b at the second surface 420b (the dashed line in FIG. 8A).

In this manner, the beveled edges 427, 421 allow the intermediate region 426b to move away from valve membrane 420 in a first direction (the direction of airflow indicated), i.e., away from the second surface 420b, e.g., as shown in FIGS. 8D and 8E, to create one or more gaps or openings 429 through the valve membrane 420 around the valve element 426 to allow airflow in the first direction. Thus, a relative positive pressure across the valve membrane 420 (adjacent the first surface 420a compared to adjacent the second surface 420b) will cause the intermediate region 426b to stretch and/or otherwise move outwardly away from the second surface 420b, while a relative negative pressure will seat the intermediate region 426b into the valve membrane 420, i.e., with the beveled edges 427 of the valve element 426 contacting and pressing against the beveled

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edges 421 of the adjacent material of the valve membrane 420, thereby preventing the valve element 426 from moving in a second direction opposite the first direction, i.e., outwardly from the first surface 420a, and preventing airflow in the second direction. Stated differently, the valve element 426 may remain substantially within a plane defined by the valve membrane 420 (i.e., substantially parallel to the first surface 412a) until a positive pressure is applied, whereupon the valve element 426 may deflect out of the plane to create the openings 429. When a negative pressure is applied, the valve element 426 may seat against the beveled edges 421 and remain substantially within the plane.

Returning to FIGS. 7A and 7B, with the valve membrane 420 of FIGS. 8A-8E attached to the base membrane 412, the valve element 426 may open to allow airflow in only one direction across the device 410. For example, with the narrower width side of the valve element 426 oriented on the same side as the adhesive, i.e., with the first surface 420a of the valve membrane 420 in FIG. 8A on the same side as the first surface 412a of the base membrane 412, the device 410 may allow airflow in the direction shown in FIG. 7B, thereby allowing a user to exhale through the device 410, while preventing inhalation.

For example, during use, the first surface 412a of the base membrane 412 may be placed over the user's mouth, e.g., after removing any backing material (not shown), such that the central open region 414 is positioned over the user's lips. Once attached, the device 410 may discourage the user to breathe through their mouth. However, in this configuration, the user may exhale through their lips and the central open region 414 (e.g., by slightly opening their lips if necessary), thereby causing the intermediate region 426b of the valve element 420 to stretch and/or move outwardly away from the second surface 420b of the valve membrane 420, as shown in FIGS. 8D and 8E. Conversely, if the user attempts to inhale through their mouth, the beveled edges 427 of the valve element 426 may be seated into the adjacent beveled edges 421, thereby substantially sealing the device 410 and preventing inhalation.

However, if the user needs to inhale, the user may forcibly separate their lips and open their mouth, which may cause the device 410 to open. For example, the large central open region 414 may allow the base membrane 412 to open inelastically, e.g., by bending or folding, or the base membrane 412 may open elastically. Given the elasticity of the material of the valve membrane 420, the valve membrane 420 may stretch, i.e., with the user separating the side edges 420d further. As the material of the valve membrane 420 stretches, gaps may be open between the beveled edges 421 of the valve membrane 420 and the valve element 426, thereby allowing the user to inhale through the gaps. When the user relaxes and/or no longer needs to inhale through their mouth, they may close their lips and the valve membrane 420 may resiliently return to its original shape with the valve element 426 seated by the beveled edges 427, 421. Alternatively, the adhesive may be configured to separate from the user's skin if the user applies sufficient force, thereby causing the device 410 to at least partially peel off and allow the user to inhale.

Alternatively, the orientation of the valve membrane 420 and valve element 426 relative to the base membrane 412 may be reversed, e.g., to substantially seal the device 410 during exhalation while allowing inhalation. Similarly for other embodiments described herein, the valve element may be oriented to permit preferential flow during inhalation or exhalation.

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Turning to FIGS. 10A and 10B, another embodiment of a valve membrane 520 is shown that may be incorporated into the breathing device 410 in FIGS. 7A and 7B instead of the valve membrane 420. In this embodiment, the valve membrane 520 includes a single valve element in the form of a flap 526 that is configured to swing open in a first direction to allow airflow in the first direction and seat to prevent airflow in a second, opposite direction. In particular, the flap 526 is defined by three slits 523 formed through the valve membrane 520 that define three beveled edges 527 with a long edge 527a extending partially between opposite ends 520c, 520d of the valve membrane 520 and short edges 527b at opposite ends of the long edge 527 such that the flap 526 remains coupled to the valve membrane 520 opposite the long edge 527a to allow the flap 526 to open in the first direction shown. Thus, a positive pressure across the valve membrane (relatively between the first and second surfaces 520a, 520a causes the flap 526 to swing open, while a relative negative pressure causes the beveled edges 527 of the flap 526 to seat against the valve membrane 520 and prevent substantial airflow. As above, the valve membrane 420 may be oriented to restrict flow during inhalation or exhalation.

Turning to FIGS. 11A-11D, another embodiment of a valve membrane 620 is shown that includes a pair of opposing valve elements 626 that open to define a gap or opening 629 between them. Similar to the valve element 426 of FIGS. 8A-8E, the valve elements 626 are defined by spaced apart longitudinal slits 623a that are formed to define beveled edges 627 along the valve elements 626. Unlike the valve element 426, a transverse slit 623b separates the valve elements 626 such that each valve element 626 defines a flap with free ends adjacent one another. As a result of this configuration, the flaps 626 may open away from the second surface 620b of the valve membrane 620 when exposed to a net positive pressure adjacent the first surface 620a without having to stretch, instead simply bending or hinging at the fixed ends, e.g., as shown in FIG. 11D. Similar to other embodiments, a net negative pressure causes the beveled edges 627 of the flaps 626 to seat against the adjacent material of the valve membrane 620, thereby preventing airflow in the opposite direction, e.g., as shown FIG. 11B.

Turning to FIG. 12, another exemplary embodiment of a breathing device 710 is shown that includes a base membrane 712 and a valve membrane 720, generally constructed similar to other embodiments herein, and configured to allow inhalation while preventing exhalation. As shown, a backing sheet 730 may be provided over an adhesive layer (not shown) on a first surface 720a thereof that may be removed before the device 710 is applied over a user's mouth. Similar to previous embodiments, the base membrane 712 includes an open central region 714 that is covered by the valve membrane 720, which includes a pair of valve elements 726.

FIG. 13 shows an alternative embodiment of a breathing device 710' similar to the device 710 of FIG. 12, except that the valve membrane 720' is oriented to allow exhalation while preventing inhalation. Similar to the device 710, the valve membrane 720' may be attached to a second surface of the base member 712,' opposite a first surface that includes an adhesive layer (not shown), similar to other embodiments herein. In this embodiment, the valve elements 726' are configured to open away from the central open region 714' (shown in phantom).

FIG. 14 shows an exemplary embodiment of a valve membrane 720 that may be provided on either of the devices 710, 710'. Similar to other embodiments, the valve mem-

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brane 720 includes a pair of flaps or valve elements 726 defined by slits 723 and including beveled edges 727 that allow the flaps 726 to open in a first direction and be seated and seal against the valve membrane 720 in a second opposite direction. The flaps 726 include an attached end and a free end. The free end comprises an arc, which is cut with a beveled edge 727. The absence of abrupt corners may allow the valve to seat and unseat easily without catching, e.g., to allow opening or closing with small changes in relative pressure across the membrane 720. However, it will be appreciated that any of the valve membranes described herein may be provided in the devices 710, 710'. It will also be appreciated that the flaps 726 may have a variety of geometries, e.g., to facilitate responsive opening and closing, e.g., with small changes in pressure differential across the membrane and effective sealing when in the closed position.

Turning to FIG. 15, optionally, in any of the embodiments herein, a backing membrane 740 may be incorporated into the breathing device to provide a back stop to support and/or limit movement of the valve element(s). For example, in the embodiment shown, the backing membrane 740 may include one or more openings 744 therethrough, e.g., corresponding and located opposite to valve element(s), such as the valve elements 726 shown in FIG. 14. The openings 744 have a width and/or height smaller than the width and/or height of the valve element(s) such that the backing membrane 740 may limit movement of the valve element(s). For example, the backing membrane 740 may be attached to a base membrane of the device on an opposite surface as the valve membrane. Thus, when the valve element(s) open through a central open region of the base membrane, the valve element(s) may contact the backing membrane 740, thereby limiting the extent to which the valve element(s) open. Alternatively, the backing membrane 740 may be positioned adjacent to the valve element(s) such that movement of the valve element(s) is limited by the backing membrane 740 and such that the valve element(s) may substantially seal against the backing membrane 740 when directed toward the backing membrane 740, e.g., by a pressure differential across the device.

Turning to FIGS. 16A and 16B, an exemplary embodiment of a breathing device 810 is shown that includes a base member 812 generally similar to other embodiments and a valve membrane 820 attached to the base member 812 that includes one or more openings 824 therethrough. As shown in FIG. 16B, a backing sheet 830 may also be provided that covers an adhesive layer on the underlying surface (not shown), also similar to other embodiments herein. Unlike other embodiments, the valve membrane 820 includes one or more openings 824 without valve elements, but having cross-sectional open area sufficient to limit airflow across the device 810 while allowing a user to breathe through the device 810 when desired or necessary.

For example, the device 810 may be placed over a user's mouth, e.g., after holding the user's lips closed together or in another relaxed position (not shown), such that the device 810 encourages nasal breathing. For example, the device 810 may hold the user's lips together to prevent mouth breathing, but allowing the user to breathe through the openings 824, if desired or necessary, e.g., if the user is startled or panics. The device 810 with openings 824 may be constructed to limit airflow in such a way that when used with nasal positive airway pressure (PAP) devices (e.g., providing pressures from about 4 to 30 cmH₂O), the resultant airflow does not generally exceed that for which an

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apparatus providing PAP can compensate, e.g., less than about 50 L/min and more ideally less than about 25 L/min.

Generally, the adhesive devices herein may be provided as a single-use device, e.g., that may be applied over a user's mouth for a single night's sleep. Thus, the adhesive device may be applied before sleep, e.g., before or after placing a PAP nasal cover over the user's nose, and then removed and discarded after waking. Alternatively, the adhesive devices may be constructed and adhesive selected to permit removal and replacement each time the user sleeps, and/or to allow other reuse.

The adhesive devices may be easily fabricated using large volume manufacturing, e.g., by providing the components as simple sheets or spooled reels of films that may be used to make multiple devices simultaneously and/or sequentially. The individual layers may be laid up and attached together and then separated into individual devices, similar to known manufacturing methods. For example, a first web may be provided for the mouth covers, e.g., having adhesive on both sides. A first side may provide an adhesive layer for later adherence to a user's mouth of an individual device, and a second side may be used to adhere to the lip closing elements and relief hole web. The optional flap may be pinned in place between the mouth cover web and the lip closing element and relief hole web. Tabs may also be included on the mouth cover web, both for easy adhesive backing removal (done by individual consumer at time of application) and/or easy of removal from a user's mouth after use is complete. Safety scores may be cut or otherwise formed into the web(s) at the same time their profile is cut from the web.

The foregoing disclosure of the exemplary embodiments has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise forms disclosed. Many variations and modifications of the embodiments described herein will be apparent to one of ordinary skill in the art in light of the above disclosure.

Further, in describing representative embodiments, the specification may have presented methods and/or processes as a particular sequence of steps. However, to the extent that the methods do not rely on the particular order of steps set forth herein, the methods should not be limited to the particular sequence of steps described. As one of ordinary skill in the art would appreciate, other sequences of steps may be possible. Therefore, the particular order of the steps set forth in the specification should not be construed as limitations on the claims.

While the invention is susceptible to various modifications, and alternative forms, specific examples thereof have been shown in the drawings and are herein described in detail. It should be understood, however, that the invention is not to be limited to the particular forms or methods disclosed, but to the contrary, the invention is to cover all modifications, equivalents and alternatives falling within the scope of the appended claims.

We claim:

1. A device for improving breathing, consisting essentially of:

- a mouth cover formed from a base membrane including first and second surfaces opposite one another, a central open region therethrough, and an adhesive on at least a portion of the first surface for detachably attaching the mouth cover to skin of a user surrounding the user's mouth;
- a removable cover sheet covering the adhesive to prevent exposure before use; and

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a valve membrane attached to one of the first and second surfaces of the base member such that the valve membrane extends across the central open region, the valve membrane including one or more valve elements adjacent the central open region, the one or more valve elements including beveled edges adjacent corresponding beveled edges of the valve membrane such that the one or more valve elements are movable away from the valve membrane in a first direction to create one or more openings through the valve membrane to allow airflow in the first direction through the central open region, the beveled edges of the valve membrane seating the one or more valve elements in the one or more openings to prevent the one or more valve elements from moving in a second direction opposite the first direction and preventing airflow in the second direction.

2. The device of claim 1, wherein the one or more valve elements comprise a valve element defined by one or more slits cut through the valve membrane to define the beveled edges.

3. The device of claim 2, wherein the valve membrane includes two slits extending substantially parallel to one another and spaced apart from one another, thereby defining the valve element between the two slits.

4. The device of claim 3, wherein the valve element comprises end regions that are substantially fixed relative to adjacent material of the valve membrane and an intermediate region that is free to move in the first direction.

5. The device of claim 4, wherein the valve membrane is formed from elastic material such that the intermediate region of the valve element stretches to move in the first direction yet is biased to return to seat beveled edges of the valve element against corresponding beveled edges of adjacent material of the valve membrane.

6. The device of claim 3, wherein the valve element comprises a first end at one end of the slits that is substantially fixed relative to adjacent material of the valve membrane and a second free end that is free to move in the first direction.

7. The device of claim 3, wherein the valve element comprises a pair of flaps opposite one another between the slits, each flap comprising a first end at one end of the slits that is substantially fixed relative to adjacent material of the valve membrane and a second free end that is free to move in the first direction.

8. The device of claim 7, wherein the second free ends are located adjacent one another at an intermediate region of the slits.

9. The device of claim 2, wherein the valve element comprises a flap valve defined by a "U" shaped slit cut through the valve membrane.

10. The device of claim 2, wherein the valve element comprises a pair of flap valves, each flap valve defined by a "U" shaped slit cut through the valve membrane.

11. The device of claim 10, wherein free ends of the flap valves are disposed closer to one another than fixed ends of the flap valves.

12. The device of claim 1, wherein the valve membrane has an elasticity greater than an elasticity of the base membrane.

13. The device of claim 1, wherein the valve membrane is formed from elastic material such that the one or more valve elements are seated within the one or more openings in a relaxed state to limit airflow in the second direction and wherein the elastic material is stretchable to open the one or

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more openings to a size larger than the one or more valve elements to allow airflow in the second direction.

14. The device of claim 1, wherein the valve membrane is attached to the first surface of the base membrane and wherein the one or more valve elements are oriented to open into the central open region to allow a user wearing the device to exhale while preventing inhalation.

15. The device of claim 1, wherein the valve membrane is attached to the second surface of the base membrane and wherein the one or more valve elements are oriented to open into the central open region to allow a user wearing the device to inhale while preventing exhalation.

16. A method for improving breathing, comprising:

applying a membrane over a user's mouth with the user's lips pressed together, the membrane comprising one or more valve elements including beveled edges such that the valve elements open freely in a first direction when the user applies a first pressure to the membrane through the user's mouth to allow airflow through the membrane in the first direction, and wherein the beveled edges seat against adjacent beveled edges of the membrane when the user applies a second pressure opposite to the first pressure to the membrane through the user's mouth to prevent substantial airflow in a second direction; and

the user separating their lips to cause the membrane to stretch such that the beveled edges of the one or more valve elements no longer seat against adjacent beveled edges of the membrane, thereby allowing the user to apply the second pressure and cause airflow in the second direction.

17. The method of claim 16, wherein the first pressure is an exhalation pressure and the second pressure is an inhalation pressure.

18. The method of claim 16, wherein the first pressure is an inhalation pressure and the second pressure is an exhalation pressure.

19. A method for improving breathing, comprising:

applying a membrane over a user's mouth with the user's lips pressed together, the membrane comprising one or more valve elements that open freely in a first direction

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when the user applies a first pressure to the membrane through the user's mouth to allow airflow through the membrane in the first direction and that close when the user applies a second pressure opposite to the first pressure to the membrane through the user's mouth to prevent substantial airflow in the second direction, and the user separating their lips to cause the membrane to stretch such that the one or more valve elements no longer close, thereby allowing the user to apply the second pressure and cause airflow in the second direction.

20. A device for improving breathing, consisting essentially of:

a base membrane including first and second surfaces opposite one another, a central open region there-through, and an adhesive on at least a portion of one of the first surface and the second surface for detachably attaching the base membrane to skin of a user surrounding the user's mouth;

a removable cover sheet over the adhesive to prevent exposure before use;

a valve membrane attached to the first surface of the base member around the central open region, the valve membrane including one or more valve elements adjacent the central open region configured such that the one or more valve elements are movable away from the base membrane in a first direction to allow airflow in the first direction through the central open region; and

a backing membrane attached to the second surface of the base membrane around the central open region and comprising one or more openings therethrough, each opening disposed opposite a corresponding one of the one or more valve elements and having an open area smaller than an area of the corresponding one of the one or more valve elements to limit airflow through the device when the corresponding one of the one or more valve elements is open, the backing member configured to prevent the one or more valve elements from moving in a second direction opposite the first direction and preventing airflow in the second direction.

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